1 Subspaces

Definition 1.1 (Subspace)**.** Let V be any vector space, and let W be a subset of V. Define vector addition and scalar multiplication on W by restricting the corresponding operations of V to W. If W is a vector space with respect to the restricted operations of V, then W is said to be a *subspace* of V.

Example 1.2. Let $V = F^2$ and $W = \{(x_1, 0) \in F^2\}$. We add and scalar multiply elements of W by using the addition and scalar multiplication of V:

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0 + 0) = (x_1 + x_2, 0)$$
$$\alpha(x_1, 0) = (\alpha x_1, 0).$$

One can check that this makes W into a subspace of F^2.

Note, in particular, that Definition 1.1 requires W to be *closed* with respect to the vector operations of V, meaning that the sum of two elements of W is in W, and the scalar multiple of an element of W is also in W. Here is an example of a subset of a vector space which is not closed with respect to the vector operations.

Example 1.3. Let $V = \mathbb{R}^2$ with the usual operations, and take $W \subseteq \mathbb{R}^2$ to be the set of points inside the unit disc; that is, $W = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$. Consider $v = (1, 0)$. Since $v + v = (2, 0) \notin W$ (since $2^2 + 0^2 = 4 > 1$), W is not closed under vector addition. Similarly, $2(1, 0) = (2, 0)$, so W is also not closed under scalar multiplication. Hence, W is not a subspace of V.

Exercise 1.4. Can the emptyset be a subspace of a vector space?

Exercise 1.5. Let W be a subspace of a vector space V.

(a) Prove that if $w \in W$, then $-w \in W$.

(b) Prove that $0 \in W$, where 0 is the zero vector of V. (*Hint*: Use part (a).)

Remark 1.6. Note from part (b) of Exercise 1.5 that if W does not contain the zero vector of V, then it cannot be a subspace of V.

Exercise 1.7. Let $V = F^2$ and let $W = \{(x_1, x_2) \in F^2 : 3x_1 + 2x_2 = 7\}$. Is W a subspace of F^2?

We have just seen that if W is a subspace of V, then it must contain the zero vector of V and be closed under addition and scalar multiplication. Conversely, suppose W is a subset of V which contains the zero vector and is closed under addition and scalar multiplication. Note that the vector space axioms, except for the existence of an additive inverse for each $w \in W$, all hold automatically in W, because they hold in V and each element of W is also an element of V (with the same operations of addition and scalar multiplication).

In part (a) of Exercise 1.5 we proved that closure of W under scalar multiplication implies that every element of W has an additive inverse in W, so in fact all of the vector space axioms hold.

Definition 1.8 (Equivalent definition of a subspace)**.** A *subspace* of a vector space V is a subset W of V which contains the zero vector and is closed under addition and scalar multiplication.

Note that this definition makes it much easier to check whether a given subset W of V is a subspace, since one does not need to check all of the vector space axioms. Here is another equivalent definition, which shows that the steps of checking for closure under addition and scalar multiplication can be combined.

Proposition 1.9 (Subspace criteria)**.** *A subset W of a vector space V is a subspace if and only if*

(i) W is non-empty, and

(ii) $\alpha \in F$ and $w_1, w_2 \in W$ implies that $\alpha w_1 + w_2 \in W$.

Exercise 1.10. Prove Proposition 1.9.

Remark 1.11. When using Proposition 1.9 to prove that a given subset W of a vector space V is a subspace, one usually shows that W is nonempty by showing that it contains the zero vector.

Exercise 1.12. Prove that every vector space V has at least two subspaces: V and $\{0\}$.

2 Examples

Example 2.1 (Subspaces of F)**.** By Exercise 1.12, $\{0\}$ and F are subspaces of F. If W is a subspace of F which contains a nonzero vector α, then W must contain every element of F, since given any $\beta \in F$, we can write $\beta = \frac{\beta}{\alpha}\alpha \in W$, since W is closed under scalar multiplication. This shows that $\{0\}$ and F are the only subspaces of F.

Exercise 2.2. By Exercise 1.12, $\{(0,0)\}$ and F^2 are subspaces of F^2. Show that

$$W := \{(x_1, x_2) \in F^2 : \alpha_1 x_1 + \alpha_2 x_2 = \beta\}$$

is a subspace of F^2 if and only if $\beta = 0$.

Remark 2.3. In the case of $\mathbb{R}^2$, the previous exercise can be interpreted geometrically as saying that a line in the plane is a subspace if and only if it passes through the origin (since the origin is the zero vector, which must be contained by a subspace). Together with the trivial subspaces ($\{0\}$ and $\mathbb{R}^2$ itself), these turn out to be the only subspaces of $\mathbb{R}^2$, though proving this will be easier after we introduce some additional tools later. We will similarly prove that the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$ itself.

Example 2.4 (Subspaces of $\mathbb{R}^{(a,b)}$)**.** Consider the vector space $\mathbb{R}^{(a,b)}$ of all real-valued functions defined on an open interval (a,b).

- Consider the set $\mathscr{C}(a,b)$ of all continuous functions on (a,b). The zero vector of $\mathbb{R}^{(a,b)}$, which is the function $0 : (a,b) \to \mathbb{R}$ which maps every element of (a,b) to 0, is continuous, so $0 \in \mathscr{C}(a,b)$. Since the sum of two continuous functions is continuous and a multiple of a continuous function by a real number is continuous, $\mathscr{C}(a,b)$ is a subspace of $\mathbb{R}^{(a,b)}$ by Proposition 1.9.

- The set $\mathscr{C}^{(n)}(a,b)$ of functions whose nth derivative exists and is continuous on (a,b), is a subspace of $\mathbb{R}^{(a,b)}$, as is the set $\mathscr{C}^{\infty}(a,b)$ of functions whose derivatives all exist and are continuous on (a,b).
- The set of all polynomials of degree n is *not* a subspace of $\mathscr{C}^{\infty}(-\infty,\infty)$ since the constant function 0 does not have degree n. However, the set of polynomials of degree $\leq n$ *is* a subspace of $\mathscr{C}^{\infty}(-\infty,\infty)$, as is the set of all polynomials. We denote these subspaces by $\mathscr{P}_n$ and $\mathscr{P}$, respectively.

3 Exercises

1. For each of the following subsets of F^3, determine whether it is a subspace of F^3
   (a) $W_1 = \{(x_1,x_2,x_3) \in F^3 : x_1 + 2x_2 + 3x_3 = 0\}$
   (b) $W_2 = \{(x_1,x_2,x_3) \in F^3 : x_1 + 2x_2 + 3x_3 = 4\}$
   (c) $W_3 = \{(x_1,x_2,x_3) \in F^3 : x_1x_2x_3 = 0\}$
   (d) $W_4 = \{(x_1,x_2,x_3) \in F^3 : x_1 = 5x_3\}$
2. Show that the set of differentiable real-valued functions f on the interval $(-4,4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.
3. Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0,1]$ such that $\int_0^1 f(x)dx = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.
4. Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?
5. (a) Is $\{(a,b,c) \in \mathbb{R}^3 : a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
   (b) Is $\{(a,b,c) \in \mathbb{C}^3 : a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?